



Accountability of AI Under the Law: The Role of Explanation

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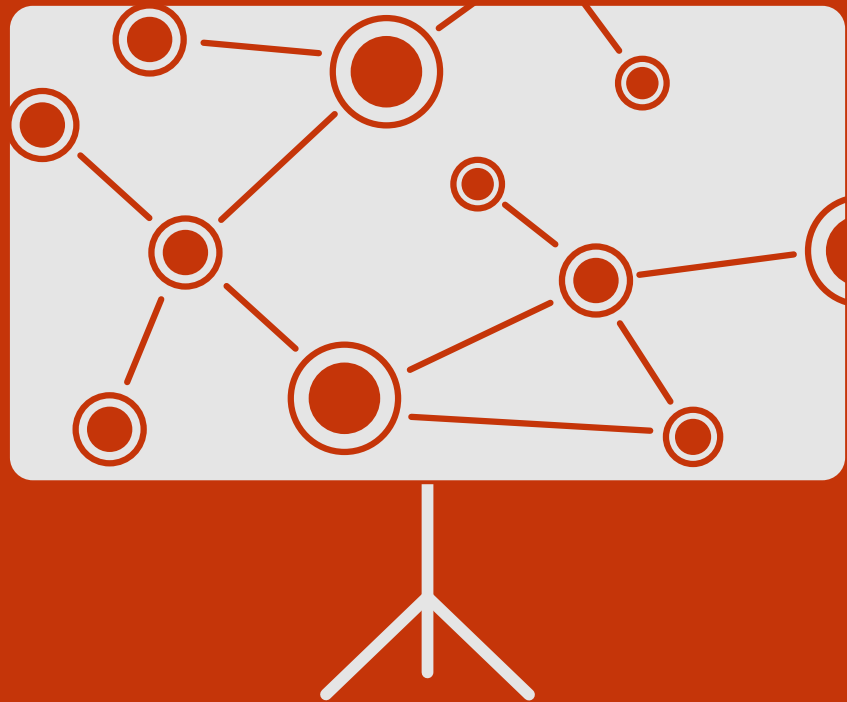
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WORKING PAPER

Accountability of AI Under the Law: The Role of Explanation

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Abstract

The ubiquity of systems using artificial intelligence or “AI” has brought increasing attention to how those systems should be regulated. The choice of how to regulate AI systems will require care. AI systems have the potential to synthesize large amounts of data, allowing for greater levels of personalization and precision than ever before—applications range from clinical decision support to autonomous driving and predictive policing. That said, our AIs continue to lag in common sense reasoning [McCarthy, 1960], and thus there exist legitimate concerns about the intentional and unintentional negative consequences of AI systems [Bostrom, 2003, Amodei et al., 2016, Sculley et al., 2014].

How can we take advantage of what AI systems have to offer, while also holding them accountable? In this work, we focus on one tool: explanation. Questions about a legal right to explanation from AI systems was recently debated in the EU General Data Protection Regulation [Goodman and Flaxman, 2016, Wachter et al., 2017a], and thus thinking carefully about when and how explanation from AI systems might improve accountability is timely. Good choices about when to demand explanation can help prevent negative consequences from AI systems, while poor choices may not only fail to hold AI systems accountable but also hamper the development of much-needed beneficial AI systems.

Below, we briefly review current societal, moral, and legal norms around explanation, and then focus on the different contexts under which explanation is currently required under the law. We find that there exists great variation around when explanation is demanded, but there also exist important consistencies: when demanding explanation from humans, what we typically want to know is whether and how certain input factors affected the final decision or outcome.

These consistencies allow us to list the technical considerations that must be considered if we desired AI systems that could provide kinds of explanations that are currently required of humans under the law. Contrary to popular wisdom of AI systems as indecipherable black boxes, we find that this level of explanation should generally be technically feasible but may sometimes be practically onerous—there are certain aspects of explanation that may be simple for humans to provide but challenging for AI systems, and vice versa. As an interdisciplinary team of legal scholars, computer scientists, and cognitive scientists, we recommend that for the present, AI systems can and should be held to a similar standard of explanation as humans currently are; in the future we may wish to hold an AI to a different standard.

1 Introduction

AI systems are currently used in applications ranging from automatic face-focus on cameras [Ray and Nicponski, 2005] and predictive policing [Wang et al., 2013] to segmenting MRI scans [Aibinu et al., 2008] and

language translation [Chand, 2016]. We expect that they will be soon be applied in safety-critical applications such as clinical decision support [Garg et al., 2005] and autonomous driving [Maurer et al., 2016]. That said, AI systems continue to be poor at common sense reasoning [McCarthy, 1960]. Thus, there exist legitimate concerns about the intentional and unintentional negative consequences of AI systems [Bostrom, 2003, Amodei et al., 2016, Sculley et al., 2014].

How can we take advantage of what AI systems have to offer, while also holding them accountable? To date, AI systems are only lightly regulated: it is assumed that the human user will use their common sense to make the final decision. However, even today we see many situations in which humans place too much trust in AI systems and make poor decisions—consider the number of car accidents due to incorrect GPS directions [Wolfe, February 17, 2014], or, at a larger scale, how incorrect modeling assumptions were at least partially responsible for the recent mortgage crisis [Donnelly and Embrechts, 2010]. As AI systems are used in more common and consequential contexts, there is increasing attention on whether and how they should be regulated. The question of how to hold AI systems accountable is important and subtle: poor choices may result in regulation that not only fails to truly improve accountability but also stifles the many beneficial applications of AI systems.

While there are many tools to increasing accountability in AI systems, we shall focus on one in this report: explanation. (We briefly discuss alternatives in Section 7.) By exposing the logic behind a decision, explanation can be used to prevent errors and increase trust. Explanations can also be used to ascertain whether certain criteria were used appropriately or inappropriately in case of a dispute. The question of when and what kind of explanation might be required of AI systems is urgent: details about a potential “right to explanation” were debated in the most recent revision of the European Union’s General Data Protection Regulation (GDPR) [Goodman and Flaxman, 2016, Wachter et al., 2017a]. While the ultimate version of the GDPR only requires explanation in very limited contexts, we expect questions around AI and explanation to be important in future regulation of AI systems—and, as noted above, it is essential that such regulation is implemented thoughtfully. In particular, there exist concerns that the engineering challenges surrounding explanation from AI systems would stifle innovation; that explanations might force trade secrets to be revealed; and that explanation would come at the price of system accuracy or other performance objective.

In this document, we first examine what kinds questions legally-operative explanations must answer. We then look at how explanations are currently used by society and, more specifically, in our legal and regulatory systems. We find that while there is little consistency about when explanations are required, there is a fair amount of consistency in what the abstract form of an explanation needs to be. This property is very helpful for creating AI systems to provide explanation; in the latter half of this document, we describe technical considerations for designing AI systems to provide explanation while mitigating concerns about sacrificing prediction performance and divulging trade secrets. Under legally operative notions of explanations, AI systems are not indecipherable black-boxes; we can, and sometimes should, demand explanation from them. We also discuss the potential costs of requiring explanation from AI systems, situations in which explanation may not be appropriate, and finally other ways of holding AI systems accountable.

This document is a product of over a dozen meetings between legal scholars, computer scientists, and cognitive scientists. Together, we are experts on explanation in the law, on the creation of AI systems, and on the capabilities and limitations of human reasoning. This interdisciplinary team worked together to recommend what kinds of regulation on explanation might be both beneficial and feasible from AI systems.

2 What is an Explanation?

In the colloquial sense, any clarifying information can be an explanation. Thus, we can “explain” how an AI makes decision in the same sense that we can explain how gravity works or explain how to bake a cake: by laying out the rules the system follows without reference to any specific decision (or falling object, or cake). When we talk about an explanation for a decision, though, we generally mean the reasons or justifications for that particular outcome, rather than a description of the decision-making process in general. In this paper, when we use the term explanation, we shall mean a human-interpretable description of the process by which

a decision-maker took a particular set of inputs and reached a particular conclusion [Wachter et al., 2017a] (see Malgieri and Comandè [2017] for a discussion about legibility of algorithmic systems more broadly).

In addition to this formal definition of an explanation, an explanation must also have the correct type of content in order for it to be useful. As a governing principle for the content an explanation should contain, we offer the following: an explanation should permit an observer to determine the extent to which a particular input was determinative or influential on the output. Another way of formulating this principle is to say that an explanation should be able to answer at least one of the following questions:

What were the main factors in a decision? This is likely the most common understanding of an explanation for a decision. In many cases, society has prescribed a list of factors that must or must not be taken into account in a particular decision. For example, we may want to confirm that a child’s interests were taken into account in a custody determination, or that race was not taken into account in a criminal prosecution. A list of the factors that went into a decision, ideally ordered by significance, helps us regulate the use of particularly sensitive information.

Would changing a certain factor have changed the decision? Sometimes, what we want to know is not whether a factor was taken into account at all, but whether it was determinative. This is most helpful when a decision-maker has access to a piece of information that has both improper and proper uses, such as the consideration of race in college admissions. By looking at the effect of changing that information on the output and comparing it to our expectations, we can infer whether it was used correctly.

Why did two similar-looking cases get different decisions, or vice versa? Finally, we may want to know whether a specific factor was determinative in relation to another decision. This information is useful when we need to assess the consistency as well as the integrity of a decision-maker. For example, it would be proper for a bank to take income into account, and even treat it as dispositive, when deciding whether to grant a loan. However, we might not want a bank to rely on income to different degrees in apparently similar cases, as this could undermine the predictability and trustworthiness of the decision-making process.

3 Societal Norms Around Explanation

Before diving into the U.S. legal context, we discuss more broadly how we, as a society, find explanations are desirable in some circumstances but not others. In doing so, we lay the foundations for specific circumstances in which explanation are (or are not) currently demanded under the law (Section 4). When it comes to human decision-makers, we often want an explanation when someone makes a decision we do not understand or believe to be suboptimal [Leake, 1992]. For example, was the conclusion accidental or intentional? Was it caused by incorrect information or faulty reasoning? The answers to these questions permit us to weigh our trust in the decision-maker and to assign blame in case of a dispute.

However, society cannot demand an explanation for every decision, because explanations are not free. Generating them takes time and effort, thus reducing the time and effort available to spend on other, potentially more beneficial conduct. Therefore, the utility of explanations must be balanced against the cost of generating them. Consider the medical profession. A doctor who explained every diagnosis and treatment plan to another doctor might make fewer mistakes, but would also see fewer patients. And so, we required newly graduated doctors to explain their decisions to more senior colleagues, but we do not require explanation from more experienced doctors—as the risk of error decreases and the value of the doctor’s time increases, the cost-benefit analysis of generating explanations shifts.

In other circumstances, an explanation might obscure more information than it reveals—humans are notoriously inaccurate when providing post-hoc rationales for decisions [Nisbett and Wilson, 1977]—and even if an explanation is accurate, we cannot ensure that it will be used in a socially responsible way. Explanations can also change an individual’s judgment: the need to explain a decision can have both positive and negative effects on the decision-maker’s choices [Messier et al., 1992], and access to an explanation might

decrease observers’ trust in some decisions [de Fine Licht, 2011]. Last but not least, social norms regarding individual autonomy weigh against demanding explanations for highly personal decisions.

What, then, are the circumstances in which the benefits of an explanation outweigh the costs? We find that there are three conditions that characterize situations in which society considers a decision-maker is obligated—morally, socially, or legally—to provide an explanation:

The decision must have been acted on in a way that has an impact on a person other than the decision maker. If a decision only impacts the decision-maker, social norms generally will not compel an explanation, as doing so would unnecessarily infringe upon the decision-maker’s independence. For example, if an individual invests their own funds and suffers losses, there is no basis to demand that the investor disclose their strategy. But if an investor makes a decision that loses a client’s money, the client may well be entitled to an explanation.

There must be value to knowing if the decision was made erroneously. Assuming the decision affects entities other than the decision-maker, society still will not demand an explanation unless the explanation can be acted on in some way. Under the law, this action usually corresponds to assigning a blame and providing compensation for injuries caused by past decisions. However, as noted in Wachter et al. [2017b], explanations can also be useful if they can positively change future decision-making. But if there is no recourse for the harm caused, then there is no justification for the cost of generating an explanation. For example, if a gambler wins a round of roulette, there is no reason to demand an explanation for the bet: there is no recourse for the casino and there is no benefit to knowing the gambler’s strategy, as the situation is not repeatable.

There must be some reason to believe that an error has occurred (or will occur) in the decision-making process. We only demand explanations when some element of the decision-making process—the inputs, the output, or the context of the process—conflicts with our expectation of how the decision will or should be made:

- **Unreliable or inadequate inputs.** In some cases, belief that an error has occurred arises from our knowledge of the decision-maker’s inputs. An input might be suspect because we believe it is logically irrelevant. For example, if a surgeon refuses to perform an operation because of the phase of the moon, society might well deem that an unreasonable reason to delay an important surgery [Margot, 2015]. An input might also be forbidden. Social norms in the U.S. dictate that certain features, such as race, gender, and sexual identity or orientation, should not be taken into account deciding a person’s access to employment, housing, and other social goods. If we know that a decision-maker has access to irrelevant or forbidden information—or a proxy for such information—it adds to our suspicion that the decision was improper. Similarly, there are certain features that we think *must* be taken into account for particular decision: if a person is denied a loan, but we know that the lender never checked the person’s credit report, we might suspect that the decision was made on incomplete information and, therefore, erroneous.
- **Inexplicable outcomes.** In other cases, belief that an error occurred comes from the output of the decision-making process, that is, the decision itself. If the same decision-maker renders different decisions for two apparently identical subjects, we might suspect that the decision was based on an unrelated feature, or even random. Likewise, if a decision-maker produces the same decision for two markedly different subjects, we might suspect that it failed to take into account a salient feature. Even a single output might defy our expectations to the degree that the most reasonable inference is that the decision-making process was flawed. If an autonomous vehicles suddenly veers off the road, despite there being no traffic or obstacles in sight, we could reasonably infer that an error occurred from that single observation.

- **Distrust in the integrity of the system.** Finally, we might demand an explanation for a decision even if the inputs and outputs appear proper because of the context in which the decision is made. This usually happens when a decision-maker is making highly consequential decisions and has the ability or incentive to do so in a way that is personally beneficial but socially harmful. For example, corporate directors may be tempted to make decisions that benefit themselves at the expense of their shareholders. Therefore, society may want corporate boards to explain their decisions, publicly and preemptively, even if the inputs and outputs of the decision appear proper [Hopt, 2011].

We observe that the question of when it is reasonable to demand an explanation is more complex than identifying the presence or absence of these three factors. Each of these three factors may be present in varying degree, and no single factor is dispositive. When a decision has resulted in a serious and plainly redressable injury, we might require less evidence of improper decision-making. Conversely, if there is a strong reason to suspect that a decision was improper, we might demand an explanation for even a relatively minor harm. Moreover, even where these three factors are absent, a decision-maker may want to voluntarily offer an explanation as a means of increasing trust in the decision-making process. To further demonstrate the complexity of determining when to requiring explanations, we now look at a concrete example: the U.S. legal system.

4 Explanations in the U.S. Legal System

The principles described in Section 3 describe the general circumstances in which we, as a society, desire explanation. We now consider how they are applied in existing laws governing human behavior. We confine our research to laws for two reasons. First, laws are concrete. Reasonable minds can and do differ about whether it is morally justifiable or socially desirable to demand an explanation in a given situation. Laws on the other hand are codified, and while one might argue whether a law is correct, at least we know what the law is. Second, the United States legal system maps well on to the three conditions from Section 3. The first two conditions—that the decision have an actual effect and that there is some benefit to obtaining an explanation—are embodied in the doctrine of standing within the constitutional injury, causation, and redressability requirements [Krent, 2001]. The third condition, reason to believe that an error occurred, corresponds to the general rule that the complaining party must allege some kind of mistake or wrongdoing before the other party is obligated to offer an explanation—in the legal system, this is called “meeting the burden of production” [Corpus Juris Secundum, c. 86 §101]. Indeed, at a high level, the anatomy of many civil cases involve the plaintiff presenting evidence of an erroneous decision, forcing the defendant to generate an innocent explanation or concede that an error occurred.

However, once we get beyond this high-level model of the legal system, we find significant variations in the demand for explanations under the law, including the role of the explanation, who is obligated to provide it, and what type or amount of evidence is needed to trigger that obligation. A few examples that highlight this variation follow:

- **Strict liability:** Strict liability is a form of legal liability that is imposed solely on the fact that the defendant caused an injury; there is no need to prove that the defendant acted wrongfully, intentionally, or even negligently. Accordingly, the defendant’s explanation for the decision to act in a certain way is irrelevant to the question of liability. Strict liability is usually based on risk allocation policies. For example, under U.S. product liability law, a person injured as a result of a poor product design decision can recover damages without reaching the question of *how* that decision was made. The intent of the strict product liability system is to place the burden of inspecting and testing products on manufacturers, who have the resources and expertise to do so, rather than consumers, who presumably do not [Owen and Davis, 2017, c. 1 §5:1].
- **Divorce:** Prior to 1969, married couples in the U.S. could only obtain a divorce by showing that one of the spouses committed some wrongful act such as abuse, adultery, or desertion—what are called “grounds for divorce.” Starting with California in 1969, changing social norms around around privacy

and autonomy, especially for women, led states to implement no-fault divorce laws, under which a couple can file for divorce without offering a specific explanation. Now, all states provide for no-fault divorce, and requiring a couple to explain their decision to separate is perceived as archaic [Guidice, 2011].

- **Discrimination:** In most discrimination cases, the plaintiff must provide some evidence that some decision made by the defendant—for example, the decision to extend a government benefit to the plaintiff—was intentionally biased before the defendant is required to present a competing explanation [Strauss, 1989]. But in certain circumstances, such as criminal jury selection, employment, or access to housing, statistical evidence that the outputs of a decision-making process disproportionately exclude a particular race or gender is enough to shift the burden of explanation on the decision-maker [Swift, 1995, Cummins and Isle, 2017]. This stems in part from the severity and prevalence of certain types of discrimination, but also a moral judgment about the repugnance of discriminating on certain characteristics.
- **Administrative decisions:** Administrative agencies are subject to different explanation requirements at different stages in their decision-making. When a new administrative policy is being adopted, the agency must provide a public explanation for the change [Corpus Juris Secundum, c. 73 §231]. But once the policies are in place, a particular agency decision is usually given deference, meaning that a court reviewing the decision will assume that the decision is correct absent countervailing evidence. Under the deferential standard, the agency only needs to show that the decision was not arbitrary or random [Corpus Juris Secundum, c. 73A §497]. Highly sensitive decisions, like national security related decisions, may be immune from any explanatory requirement at all.
- **Judges and juries:** Whether and how a particular judicial decision must be explained varies based on a number of factors, including the importance of the decision and the nature of the decision-maker. For example, a judge ruling on a motion to grant a hearing can generally do so with little or no explanation; the decision is highly discretionary. But a judge handing down a criminal sentence—one of the most important decisions a court can make—must provide an explanation so that the defendant can detect and challenge any impropriety or error [O’Hear, 2009]. On the other hand, a jury cannot be compelled to explain why it believed a certain witness or drew a certain inference, even though these decisions may have an enormous impact on the parties. One justification given for not demanding explanations from juries is that public accountability could bias jurors in favor of making popular but legally incorrect decisions; another is that opening jury decisions to challenges would weaken public confidence in the outcomes of trials and bog down the legal system [Landsman, 1999].

As the foregoing examples show, even in the relatively systematic and codified realm of the law, there are numerous factors that affect whether human decision-makers will be required to explain their decisions. These factors include the nature of the decision, the susceptibility of the decision-maker to outside influence, moral and social norms, the perceived costs and benefits of an explanation, and a degree of historical accident.

5 Implications for AI systems

With our current legal contexts in mind, we now turn to technical considerations for extracting explanation from AI systems. That is, how challenging would it be to create AI systems that provide the same kinds of explanation that are currently expected of humans, in the contexts that are currently expected of humans, under the law? Human decision-makers are obviously different from AI systems (see Section 6 for a comparison), but in this section we answer this question largely in the affirmative: for the most part, it *is* technically feasible to extract the kinds of explanations that are currently required of humans from AI systems.

Legally-Operative Explanations are Feasible. The main source of this feasibility arises from the fact that explanation is *distinct* from transparency. Explanation does not require knowing the flow of bits through

an AI system, no more than explanation from humans requires knowing the flow of signals through neurons (neither of which would be interpretable to a human!). Instead, explanation, as required under the law, as outlined in Section 2, is about answering how certain factors were used to come to the outcome in a specific situation. These core needs can be formalized by two technical ideas: *local explanation* and *local counterfactual faithfulness*.

Local Explanation. In the AI world, explanation for a specific decision, rather than an explanation of the system’s behavior overall, is known as local explanation [Ribeiro et al., 2016, Lei et al., 2016, Adler et al., 2016, Fong and Vedaldi, 2017, Selvaraju et al., 2016, Smilkov et al., 2017, Shrikumar et al., 2016, Kindermans et al., 2017, Ross et al., 2017, Singh et al., 2016]. AI systems are naturally designed to have their inputs varied, differentiated, and passed through many other kinds of computations—all in a reproducible and robust manner. It is already the case that AI systems are trained to have relatively simple decision boundaries to improve prediction accuracy, as we do not want tiny perturbations of the input changing the output in large and chaotic ways [Drucker and Le Cun, 1992, Murphy, 2012]. Thus, we can readily expect to answer the first question in Section 2—what were the important factors in a decision—by systematically probing the inputs to determine which have the greatest effect on the outcome. This explanation is *local* in the sense that the important factors may be different for different instances. For example, for one person, payment history may be the reason behind their loan denial, for another, insufficient income.

Counterfactual Faithfulness. The second property, counterfactual faithfulness, encodes the fact that we expect the explanation to be causal. Counterfactual faithfulness allows us to answer the remaining questions from Section 2: whether a certain factor determined the outcome, and related, what factor caused a difference in outcomes. For example, if a person was told that their income was the determining factor for their loan denial, and then their income increases, they might reasonably expect that the system would now deem them worthy of getting the loan. Importantly, however, we only expect that counterfactual faithfulness apply for related situations—we would not expect an explanation in a medical malpractice case regarding an elderly, frail patient to apply to a young oncology patient. However, we may expect it to still hold for a similar elderly, less frail patient. Recently Wachter et al. [2017b] also point out how counterfactuals are the cornerstone of what we need from explanation.

Importantly, both of these properties above can be satisfied *without* knowing the details of how the system came to its decision. For example, suppose that the legal question is whether race played an inappropriate role in a loan decision. One might then probe the AI system with variations of the original inputs changing only the race. If the outcomes were different, then one might reasonably argue that gender played a role in the decision. And if it turns out that race played an inappropriate role, that constitutes a legally sufficient explanation—no more information is needed under the law (although the company may internally choose decide to determine the next level of cause, e.g. bad training data vs. bad algorithm). This point is important because it mitigates concerns around trade secrets: explanation can be provided without revealing the internal contents of the system.

Explanation systems should be considered distinct from AI systems. We argue that regulation around explanation from AI systems should consider the explanation system as *distinct* from the AI system. Figure 1 depicts a schematic framework for explainable AI systems. The AI system itself is a (possibly proprietary) black-box that takes in some inputs and produces some predictions. The designer of the AI system likely wishes the predictions ($\hat{y}$) to match the real world (y). The designer of the *explanation system* must output a *human-interpretable* rule $e_x()$ that takes in the same input x and outputs a prediction $\tilde{y}$. To be locally faithful under counterfactual reasoning formally means that the predictions $\tilde{y}$ and $\hat{y}$ are the same under small perturbations of the input x .

This framework renders concepts such as local explanation and local counterfactual faithfulness readily quantifiable. For any input x , we can check whether the prediction made by the local explanation ($\tilde{y}$) is the same as the prediction made by the AI system ($\hat{y}$). We can also check whether these predictions remain consistent over small perturbations of x (e.g. changing the race). Thus, not only can we measure what proportion of the time an explanation system is faithful, but we can also identify the specific instances in which it is not. From a regulatory perspective, this opens the door to regulation that requires that an AI

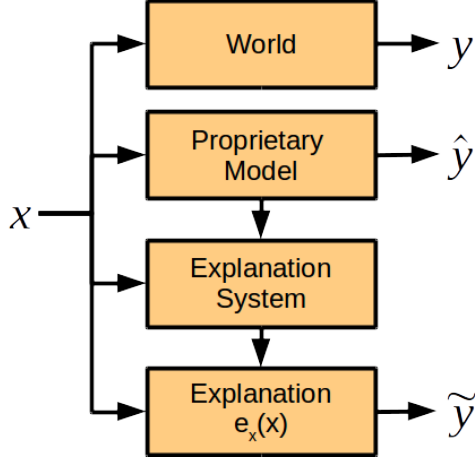


Figure 1: Diagram of a Framework for Explainable AI Systems.

system be explainable some proportion of the time or in certain kinds of contexts—rather than all the time. Loosening the explanation requirement in this way may allow for the AI system to use a much more complex logic for a few cases that really need it. More broadly, thinking of an explanation system as distinct from the original AI system also creates opportunities for industries that specialize in explanation systems.

There will exist challenges in mapping inputs and intermediate representations in AI systems to human-interpretable concepts. While the notion of how explanations are used under the law can be formalized computationally, there remains a key technical challenge of converting the inputs to an AI system—presumably some large collection of variables, such as pixel values—into human-interpretable terms such as age or gender. For example, self-driving cars may have multitudes of sensors, each with high-dimensional range and vision inputs; the human brain already converts its visual inputs into higher-level concepts such as trees or street signs. Clinical decision support systems may take in tens of thousands of variables about a patient’s diagnoses, drugs, procedures, and concepts extracted from the clinical notes; the human doctor has terms like sepsis or hypertension to describe constellations of these variables. While there do exist methods to map the high-dimensional inputs to an AI system to human-interpretable concepts, the process generally requires training the system with large amounts of data in which both the raw input and the associated concept are given.

As such, explanations from AI systems will be most straight-forward if the relevant terms are known in advance. In this case, the AI system can be trained to map its inputs to the relevant terms. For example, in the medical sphere, there are a number of algorithms for determining whether a patient has diabetes from a multitude of inputs [Newton et al., 2013]; recent work has identified ways to weigh the importance of much more general terms [Been Kim]. There will be some technical innovation required, but by and large we see relatively few difficulties for AI systems to provide the kinds of explanation that are currently required in the case where legislation or regulation makes it clear what terms may be asked for *ex ante*; there is also an established process for companies to adapt new standards as legislation and regulation change. That said, there are subtleties. While it is relatively straightforward to identify what inputs are correlated with certain terms, and verify whether predictions of terms are correlated with decisions, it will require some work to determine ways to test counterfactuals. For example, how can we show that a security system that uses images of a face as input does not discriminate against gender? One would need to consider an alternate face that was similar in every way except for gender.

Another subtlety is that, to create the required terms, the AI system will need access to potentially sensitive information. Currently, we often assume that if the human did not have access to a particular term, such as race, then it could not have been used in the decision. However, it is very easy for AI systems

to reconstruct sensitive terms from high-dimensional inputs. Data about shopping patterns can be used to identify term such as age, gender, and socio-economic status, as can data about healthcare utilization. Especially with AI systems, excluding a protected category does not mean that a proxy for that category is not being created. Thus, a corollary to the arguments above is that we must measure any terms that we wish to protect against, to be able to ensure that we are not generating proxies for them. Our legal system must allow them to be collected, and AI system designers should build ways to test whether systems are creating that term and using it inappropriately. Regulation must be put in place so that any protected terms collected by AI system designers are used only to ensure that the AI system is designed correctly, and not for other purposes within the organization. (It would be unfortunate, to say the least, if we can verify that an AI system is not discriminating against a protected term, only to find that a human decision-maker is accessing and combining the forbidden information with the AI system’s recommendation to make a final choice.)

The challenges increase if the relevant terms cannot be determined in advance. For example, in litigation scenarios, the list of relevant terms is generally only determined *ex post*. In such cases, AI systems may struggle; unlike humans, they cannot be asked to refine their explanations after the fact without additional training data. For example, we cannot identify what proxies there are for age in a data set if age itself has never been measured. For such situations, we first note that there is precedent for what to do in litigation scenarios when some information is not available, ranging from drawing inferences against the party that could have provided the information to imposing civil liability for unreasonable record-keeping practices [Nolte, 1994, Cicero, 1988]. Second, while not always possible, in many cases it may be possible to quickly train a proxy—especially if AI designers have designed the system to be updated—or have the parties mutually agree (perhaps via a third party) what are acceptable proxies. The parties may also agree to assessment via non-explanation-based tools.

In summary, to build AI systems that can provide explanation in terms of human-interpretable terms, we must both list those terms and allow the AI system access to examples to learn them. System designers should design systems to learn these human-interpretable terms, and also store data from each decision so that is possible to reconstruct and probe a decision post-hoc if needed. Policy makers should develop guidelines to ensure that the explanation system is being faithful to the original AI.

6 A Comparison of Human and AI Capability for Explanation

So far, we have argued that explanation from AI is technically feasible in many situations. However, there are obviously salient differences between AI systems and humans. Should this affect the extent to which AI explanations should be the subject of regulation? We begin with the position that, in general, AIs should be capable of providing an explanation in any situation where a human would be legally required to do so. This approach would prevent otherwise legally accountable decision-makers from “hiding” behind AI systems, while not requiring the developers of AI systems to spend resources or limit system performance simply to be able to generate legally unnecessary explanations.

That said, given the differences between human and AI processes, there may be situations in which it is possible to demand more from humans, and other situations in which it might be possible to hold AI systems to a higher standard of explanation. There are far too many factors that go into determining when an explanation should be legally required to analyze each of them with respect to both humans and AIs in this paper. At the most general level, though, we can categorize the factors that go into such a determination as either extrinsic or intrinsic to the decision-maker. Extrinsic factors—the significance of the decision, the relevant social norms, the extent to which an explanation will inform future action—are likely to be the same whether the decision-maker is a human or an AI system.

Intrinsic factors, though, may vary significantly between humans and AIs (see Table 1), and will likely be key in eventually determining where demands for human and AI explanations under the law should overlap and where they should diverge. One important difference between AIs and humans is the need to pre-plan explanations. We assume that humans will, in the course of making a decision, generate and store the information needed to explain that decision later if doing so becomes useful. A doctor who does not

explain the reasons for a diagnosis at the time it is made can nevertheless provide those reasons after the fact if, for example, diagnosis is incorrect and the doctor gets sued. A decision-maker might be required to create a record to aid in the subsequent generation of an explanation—to continue the prior example, many medical providers require doctors to annotate patient visits for this very reason, despite the fact that it takes extra time. However, requiring human decision-makers to document their decisions is the exception, not the norm. Therefore, the costs and benefits of generating an human explanation can be assessed at the time the explanation is requested.

In contrast, AI systems do not automatically store information about their decisions. Often, this feature is considered an advantage: unlike human decision-makers, AI systems can delete information to optimize their data storage and protect privacy. However, an AI system designed this way would not be able to generate *ex post* explanations the way a human can. Instead, whether resources should to be allocated to explanation generation becomes a question of system design. This is analogous to the question of whether a human decision-maker should be required to keep a record. The difference is that with an AI system this design question must *always* be addressed explicitly.

That said, AI systems can be designed to store their inputs, intermediate steps, and outputs exactly (although transparency may be required to verify this). Therefore, they do not suffer from the cognitive biases that make human explanations unreliable. Additionally, unlike humans, AI systems are not vulnerable to the social pressures that could alter their decision-making processes. Accordingly, there is no need to shield AI systems from generating explanations, for example, the way the law shields juries.

Table 1: Comparison of Human and AI Capabilities for Explanation

	<i>Human</i>	<i>AI</i>
<i>Strengths</i>	Can provide explanation post-hoc	Reproducible, no social pressure
<i>Weaknesses</i>	May be inaccurate and unreliable, feel social pressure	Requires up-front engineering, explicit taxonomies and storage

7 Alternatives to Explanation

Explanation is but one tool to hold AI systems accountable. In this section, we discuss the trade-offs associated with three core classes of tools: explanation, empirical evidence, and theoretical guarantees.

Explanation. In Section 5, we noted that an explanation system may struggle if a new factor is suddenly needed. In other cases, explanation may be possible but undesirable for other reasons: Designing a system to also provide explanation is a non-trivial engineering task, and thus requiring explanation all the time may create a financial burden that disadvantages smaller companies; if the decisions are low enough risk, we may not wish to require explanation. In some cases, one may have to make trade-offs between the proportion of time that explanation can be provided and the accuracy of the system; that is, by requiring explanation we might cause the system to reject a solution that cannot be reduced to a human-understandable set of factors. Obviously, both explanation and accuracy are useful for preventing errors, in different ways. If the overall number of errors is lower in a version of the AI system that does not provide explanation, then we might wish to only monitor the system to ensure that the errors are not targeting protected groups and the errors even out over an individual. Similar situations may occur even if the AI is not designed to reject solutions that fall below a threshold of explicability; the human responsible for implementing the solution may discard it in favor of a less optimal decision with a more appealing—or legally defensible—explanation. In either case, society would lose out on an optimal solution. Given that one of the purported benefits of AI decision-making is the ability to identify patterns that humans cannot, this would be counterproductive.

Empirical Evidence. Another tool for accountability is empirical evidence, that is measures of a system’s overall performance. Empirical evidence may justify (or implicate) a decision-making system by demonstrat-

ing the value (or harm) of the system, without providing an explanation for any given decision. For example, we might observe that an autonomous aircraft landing system has fewer safety incidents than human pilots, or that the use of a clinical diagnostic support tool reduces mortality. Questions of bias or discrimination can be ascertained statistically: for example, a loan approval system might demonstrate its bias by approving more loans for men than women when other factors are controlled for. In fact, in some cases statistical evidence is the only kind of justification that is possible; certain types of subtle errors or discrimination may only show up in aggregate. While empirical evidence is not unique to AI systems, AI systems, as digesters of data used in highly reproducible ways, are particularly well-suited to provide empirical evidence. However, such evidence, by its nature, cannot be used to assign blame or innocence surrounding a particular decision.

Theoretical Guarantees. In rarer situations, we might be able to provide theoretical guarantees about a system. For example, we trust our encryption systems because they are backed by proofs; neither explanation or evidence are required. Similarly, if there are certain agreed-upon schemes for voting and vote counting, then it may be possible to design a system that provably follows those processes. Likewise, a lottery is shown to be fair because it abides by some process, even though there is no possibility of fully explaining the generation of the pseudo-random numbers involved. Theoretical guarantees are a form of perfect accountability that only AI systems can provide, and ideally will provide more and more often in the long term; however, these guarantees require very cleanly specified contexts that often do not hold in real-world settings.

We emphasize that the trade-offs associated with all of these methods will shift as technologies change. For example, access to greater computational resources may reduce the computational burden associated with explanation, but enable even more features to be used, increasing the challenges associated with accurate summarization. New modes of sensing might allow us to better measure safety or bias, allowing us to rely more on empirical evidence, but they might also result in companies deciding to tackle even more ambitious, hard-to-formalize problems for which explanation might be the only available tool. We summarize considerations for choosing an accountability tool for AI systems in Table 2.

Table 2: Considerations for Approaches for Holding AIs Accountable

<i>Approach</i>	<i>Well-suited Contexts</i>	<i>Poorly-suited Contexts</i>
Theoretical Guarantees	Situations in which both the problem and the solution can be fully formalized (gold standard, for such cases)	Any situation that cannot be sufficiently formalized (most cases)
Statistical evidence	Problems in which outcomes can be completely formalized, and we take a strict liability view; problems where we can wait to see some negative outcomes happen so as to measure them	Situations where the objective cannot be fully formalized in advance
Explanation	Problems that are incompletely specified, where the objectives are not clear and inputs might be erroneous	Situations in which other forms of accountability are not possible

8 Recommendations

In the sections above, we have discussed the circumstances in which humans are required to provide explanation under the law, as well as what those explanations are expected to contain. We have also argued that

it should be technically feasible to create AI systems that provide the level of explanation that is currently required of humans. The question, of course, is whether we *should*. The fact of the matter is that AI systems are increasing in capability at an astounding rate, with optimization methods of black-box predictors that far exceed human capabilities. Making such quickly-evolving systems be able to provide explanation, while feasible, adds an additional amount of engineering effort that might disadvantage less-resourced companies because of the additional personnel hours and computational resources required; these barriers may in turn result in companies employing suboptimal but easily-explained models.

Thus, just as with requirements around human explanation, we will need to think about why and when explanations are useful enough to outweigh the cost. Requiring every AI system to explain every decision could result in less efficient systems, forced design choices, and a bias towards explainable but suboptimal outcomes. For example, the overhead of forcing a toaster to explain why it thinks the bread is ready might prevent a company from implementing a smart toasting feature—either due to the engineering challenges or concerns about legal ramifications. On the other hand, we may be willing to accept the monetary cost of an explainable but slightly less accurate loan approval system for the societal benefit of being able to verify that it is nondiscriminatory. As discussed in Section 3, there are societal norms around when we need explanation, and these norms should be applied to AI systems as well.

For now, we posit that demanding explanation from AI systems in such cases is not so onerous that we should ask of our AI systems what we ask of humans. Doing so avoids AI systems from getting a “free pass” to avoid the kinds of scrutiny that may come to humans, and also avoids asking so much of AI systems that it would hamper innovation and progress. Even this modest step will have its challenges, and as they are resolved, we will gain a better sense of whether and where demands for explanation should be different between AI systems and humans. As we have little data to determine the actual costs of requiring AI systems to generate explanations, the role of explanation in ensuring accountability must also be re-evaluated from time to time, to adapt with the ever-changing technology landscape.

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